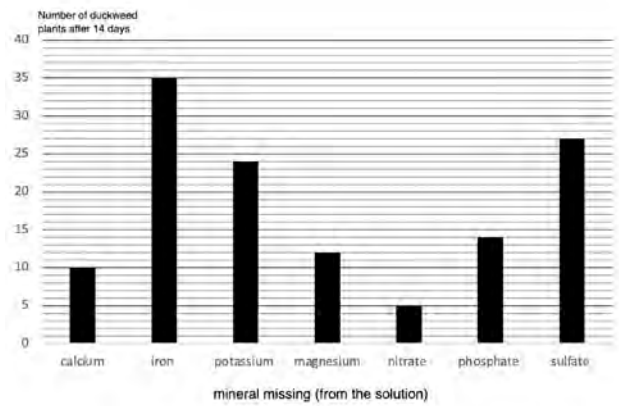


Question Number	Answer	Additional Guidance	Mark
2(a)(i)	An answer that includes the following points: • (one duckweed plant and) a solution with all minerals (1) • (one duckweed plant and) a solution with no minerals / (distilled) water (1)	accept complete solution/medium	(2)

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	A description that includes the following points: - temperature by use of {thermostatically controlled {chamber / room} / incubator} (1) - pH by use of buffer (1)		(2)

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	A graph showing the following features: - A axes correct (x – mineral missing, y – no. of plants), y starting at zero and with no break in the axis (1) - L axes correctly labelled (1) - P correct plotting on a linear scale on y (1) - S bar chart (1)	 accept minerals in any order on x	(4)

Question Number	Answer	Additional Guidance	Mark
2(b)(ii)	An explanation that includes the following points: • fewer duckweed plants with no nitrate than with no phosphate / less growth with no nitrate than with no phosphate (1) • credit correct use of nitrate (1) • further detail of role of the nitrate containing substance stated in mp2 (1)	accept nitrate least growth piece together e.g amino acids/protein/chlorophyll/ATP/nucleic acids/chlorophyll/enzymes e.g. protein enzymes / energy / ATP / cell division / new plant/ photosynthesis/respiration/speed up reactions	(3)

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	An answer that includes two of the following points: • difficult to distinguish one plant from another/plants overlap (1) • (plants / leaflets) are of different sizes / may grow by increasing in size (1) • leaf number varies from plant to plant (1)		(2)

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	An answer that includes the following points: • measure the {mass/area/root length} (1) • measurements of growth taken (at start) and after {stated/known/intervals of} time • description of method to improve accuracy of measurement of growth (1) • calculation of rate as {change divided by time / gradient of graph against time}	accept weight ecf eg. height 1 day minimum if stated e.g. pat dry before weighing, {2 or higher place / electronic} balance, use of calipers, graph paper, micrometer	(4)